

991 + 981 , MY 2012 =>

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Diagnosis information - Front-end electronics

Front-end electronics, Production mode Active

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Production mode Active

Fault effects

- ♦ Functions in the control unit are available only with restrictions.

Diagnosis information - Front-end electronics

no Engine oil pressure

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

Front-end electronics No function

Diagnosis information - Front-end electronics

Rpm signal PSM control module, implausible

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- **Vehicle speed** $v > 10 \text{ km/h}$
- **Supply voltage** $7 < u < 18 \text{ V}$

Fault setting condition(s)

- **Wheel speed**: Implausible signal from **PSM control module** for $t > 10 \text{ s}$

Fault effects

- **Electronic steering column lock (ELV)** No function

INFORMATION

- **Electronic steering column lock (ELV)** does not lock if Fault is entered passively

Diagnosis information - Front-end electronics

Crash signal Airbag control unit, implausible

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

INFORMATION

Transmission of the crash signal (PWM) by the **Airbag control unit** via signal line to the **Control module Front-end electronics** and **DME control module**

WARNING

Uncontrolled triggering of the airbags

- **Contusions, crushing injuries or death**

- ▶ Before working on the ignition circuits of the airbag system, always disconnect the battery from the vehicle electrical system and ensure that the vehicle electrical system is de-energized.
- ▶ Wait for at least one minute before starting work on the airbag system.
- ▶ Perform fault finding on the ignition circuits of the airbag system and its lines only using approved special tools.

Possible fault causes

- ♦ Short circuit to B+/ground or open circuit in the line
- ♦ Loose contact or corrosion on connectors or lines
- ♦ **Airbag control unit** Faulty
- ♦ **DME control module** Faulty

Fault setting condition(s)

- ♦ **Crash signal from Airbag control unit implausible**

80300B

Diagnosis information - Front-end electronics

Ignition lock Open circuit

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

Front-end electronics No function

Diagnosis information - Front-end electronics

Filler flap Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Open circuit/short circuit to ground
- **Filler-flap servo motor** blocked

Fault effects

- ♦ **Filler flap** No function

Diagnosis information - Front-end electronics

Horn Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Open circuit/short circuit to ground
- **Horn** Faulty

Fault effects

- ♦ **Horn** No function

Diagnosis information - Front-end electronics

Servo motor for lid release faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

- ♦ Lid does not function

Diagnosis information - Front-end electronics

Front-end electronics Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

- **Front-end electronics** No function

Diagnosis information - Front-end electronics

Steering Limp-home

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

- ♦ Steering Limp-home

Diagnosis information - Front-end electronics

Front-end electronics Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault effects

- **Front-end electronics** No function

Diagnosis information - Front-end electronics

Steering Limp-home

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

- ♦ Steering Limp-home

Diagnosis information - Front-end electronics

Steering Valve

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on
- Short circuit to B+/ground/between lines or open circuit in the lines

Fault effects

- ♦ Steering Limited function

Diagnosis information - Front-end electronics

Steering Valve

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on
- Short circuit to B+/ground/between lines or open circuit in the lines

Fault effects

- ♦ Steering Limited function

Diagnosis information - Front-end electronics

Steering Valve

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on
- Short circuit to B+/ground/between lines or open circuit in the lines

Fault effects

- ♦ Steering Limited function

Diagnosis information - Front-end electronics

Steering Valve

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on
- Short circuit to B+/ground/between lines or open circuit in the lines

Fault effects

- ♦ Steering Limited function

Diagnosis information - Front-end electronics

DME control module Confirmation failed

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Different data **DME control module** <-> **Front-end electronics** during teaching
- **Front-end electronics** incomplete taught

► Teach >component protection< using >PIWIS Tester II< ⇒ >Diagnostics< ⇒ >Online teaching functions<.

Fault effects

- ♦ no Engine running

Diagnosis information - Front-end electronics

Electronic steering column lock (ELV) Confirmation failed

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Implausible signal during **Electronic steering column lock (ELV)** locked

/ unlocked

Fault effects

- ♦ **Electronic steering column lock (ELV)** locked

/ unlocked not

Possible fault causes

- ♦ **Electronic steering column lock (ELV)** not to Immobilizer taught

Diagnosis information - Front-end electronics

Rear-end electronics Confirmation failed

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Different data **DME control module** <-> **Rear-end electronics** during teaching
- **Front-end electronics** incomplete taught

Fault effects

- ♦ Alarm system triggers

Diagnosis information - Front-end electronics

High-voltage power electronics control unit Confirmation failed

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Different data **High-voltage power electronics control unit** <-> **Front-end electronics** during teaching
- **Front-end electronics** incomplete taught

Fault effects

- ♦ no Engine start

80530F

Diagnosis information - Rear-end electronics

NOTE

For vehicles with **Porsche Vehicle Tracking System (PVTS)**, see **Tequipment (TEQ) instructions**, main group **9063**!

► This fault code relates to the **Porsche Vehicle Tracking System (PVTS)**.
More information is available in attached documents or from the respective service provider.

Diagnosis information - Front-end electronics

Immobilizer not taught

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Immobilizer not taught

Fault effects

- ♦ Engine start **not** possible

Diagnosis information - Front-end electronics

Garage door opener Faulty

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on
- **Garage door opener** in the **Control module Front-end electronics** coded

Fault setting condition(s)

- **Garage door opener** sends fault signal to **Control module Front-end electronics**

Fault effects

- ♦ **Garage door opener** No function

Diagnostic information - front-end electronics

Rain/light sensor, voltage

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, clear the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Above limit value
- Below limit value

Fault effects

Rain/light sensor, no function

Diagnostic information - front-end electronics

Rain/light sensor, voltage

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, clear the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Fault setting condition(s)

- Above limit value
- Below limit value

Fault effects

Rain/light sensor, no function

Diagnosis information - Front-end electronics

Sensor faulty

Diagnostic conditions

- The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

- Sensor does not function

Diagnosis information - Front-end electronics

Sensor faulty

Diagnostic conditions

- The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.
In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

- Sensor does not function

Diagnosis information - Front-end electronics

Rain sensor no/incorrect coding/calibration

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on
- **Rain sensor** coded

Fault effects

Rain sensor No function

Diagnosis information - Front-end electronics

Rain sensor Implausible signal

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on
- **Rain sensor** coded

Fault effects

Rain sensor No function

Diagnosis information - Front-end electronics

Electrically adjustable steering column (EVLS)

⚠ CAUTION

Adjustment of the **Electrically adjustable steering column (EVLS) !**
Intervention during the adjustment can cause injuries to the hands or fingers.

♦ Switch the ignition off or disconnect the power supply for the **Electrically adjustable steering column (EVLS)** .

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

Electrically adjustable steering column (EVLS) No function

Diagnosis information - Front-end electronics

Electrically adjustable steering column (EVLS)

⚠ CAUTION

Adjustment of the **Electrically adjustable steering column (EVLS) !**
Intervention during the adjustment can cause injuries to the hands or fingers.

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Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

Electrically adjustable steering column (EVLS) No function

Diagnosis information - Front-end electronics

Electrically adjustable steering column (EVLS)

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Diagnostic conditions

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In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

Electrically adjustable steering column (EVLS) No function

Diagnosis information - Front-end electronics

Electrically adjustable steering column (EVLS)

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♦ Switch the ignition off or disconnect the power supply for the **Electrically adjustable steering column (EVLS)** .

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

Electrically adjustable steering column (EVLS) No function

Diagnosis information - Front-end electronics

Electrically adjustable steering column (EVLS)

⚠ CAUTION

Adjustment of the **Electrically adjustable steering column (EVLS) !**
Intervention during the adjustment can cause injuries to the hands or fingers.

♦ Switch the ignition off or disconnect the power supply for the **Electrically adjustable steering column (EVLS)** .

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

Electrically adjustable steering column (EVLS) No function

Diagnosis information - Front-end electronics

Switch Electrically adjustable steering column (EVLS)

⚠ CAUTION

Adjustment of the **Electrically adjustable steering column (EVLS) !**
Intervention during the adjustment can cause injuries to the hands or fingers.

♦ Switch the ignition off or disconnect the power supply for the **Electrically adjustable steering column (EVLS)** .

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

Fault setting condition(s)

- Ignition on

Fault effects

Electrically adjustable steering column (EVLS) No function

Diagnosis information - Front-end electronics

Electrically adjustable steering column (EVLS) no/incorrect coding/calibration

▲ **CAUTION**

Adjustment of the **Electrically adjustable steering column (EVLS) !**
Intervention during the adjustment can cause injuries to the hands or fingers.

♦ Switch the ignition off or disconnect the power supply for the **Electrically adjustable steering column (EVLS)** .

Diagnostic conditions

The control unit must be active in order to test internal function sequences and diagnosable inputs/outputs.

In the event of a fault, check whether the fault is permanently >present<. To perform the check, erase the fault memory, meet the diagnostic conditions and read out the fault memory again.

- Ignition on

Possible fault causes

- ♦ **Electrically adjustable steering column (EVLS)** not coded
- ♦ **Electrically adjustable steering column (EVLS)** not calibrated

Fault effects

Electrically adjustable steering column (EVLS) No function